

Applying algorithms

Apart from computer science, algorithms can also be applied to real-life situations. There is often more than one algorithm for solving any particular problem.

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Algorithms

Algorithms can be made to do many things, from sorting a list to finding the fastest route between two locations. They can even work out the best strategies for playing games. Every programming language implements the same algorithm differently, though the end results are the same.

▷ Algorithms vs programs

An algorithm can be compared to a map and a program to a city. While algorithms focus only on logic, programs contain language-specific details and syntax.



A map shows only essential details like streets and buildings.

In reality the city has streets and buildings in full 3D detail.

Algorithm efficiency

Two different roads might lead to the same place, but taking the motorway is faster than a winding mountain path. This is true for algorithms as well. Two algorithms might produce the same result, but one may be more efficient than the other.

▽ Adding numbers

If you were asked to find the sum of all the numbers from 1 to 100, you could add them one-by-one. German mathematician Carl Gauss worked out a much quicker and more efficient way to do it that required just two simple steps.

$$1 + 2 + 3 + \dots + 100 = ?$$

That's simple! The answer is $50 \times 101 = 5050$.



REAL WORLD

Self-driving car

A self-driving car does the same tasks as a human driver: navigate lanes, detect stop signs, and respond to traffic lights. To make self-driving cars safe and reliable, programmers spend a lot of time writing the perfect algorithm for each task. Efficiency is crucial. Otherwise, the car might only detect the stop sign after it's driven past!

